

Irreducibility of an arithmetic-progression polynomial

Theorem 1. *For every integer $n \geq 2$, the polynomial*

$$f_n(x) = \sum_{k=1}^n kx^{n-k} = x^{n-1} + 2x^{n-2} + \cdots + (n-1)x + n$$

is irreducible in $\mathbb{Q}[x]$ and in $\mathbb{Z}[x]$.

The argument is by contradiction. If a factorization existed, certain products of differences between roots would be nonzero integers. An exact identity would force their product to be large. We bound these integers in two ways: using ordinary distances between complex roots, and using the powers of primes that divide them. The bounds conflict.

The main comparison first rules out factorizations whose two constant terms are at least 13. Such a factorization would have $n \geq 182$. The final section excludes smaller constant terms, completing the proof also for $2 \leq n < 182$. The first stage requires the condition on the constant terms; $n \geq 182$ alone is not enough.

We assume polynomial arithmetic, complex numbers, elementary linear algebra, and calculus. The products called discriminants and resultants are defined below. The local-field theorems needed for the prime-by-prime argument go beyond a usual undergraduate course; their precise working forms are stated before use, with a reference for their proofs.

Throughout, $N = n + 1$, $\ln$ denotes the natural logarithm, and $\text{rad}(m)$ is the product of the distinct prime divisors of a positive integer m .

1 Root estimates

We first locate the roots in a thin annulus just outside the unit circle. The first lemma is used throughout. The next two give stronger lower bounds needed in the numerical comparison and in the final section.

Direct multiplication gives

$$(x-1)f_n(x) = \sum_{j=1}^n x^j - n, \quad F(x) := (x-1)^2 f_n(x) = x^N - Nx + n. \quad (1)$$

Also $f_n(1) = nN/2$, so the root 1 of F has multiplicity exactly two.

Lemma 2 (The root annulus). *Every root α of f_n satisfies*

$$1 < |\alpha| < (2n+1)^{1/n}.$$

Proof. If $|\alpha| \leq 1$, the first identity in (1) gives

$$n = \left| \sum_{j=1}^n \alpha^j \right| \leq \sum_{j=1}^n |\alpha|^j \leq n.$$

Equality requires $|\alpha| = 1$ and all the summands α^j to point in the direction of their positive real sum. In particular α is positive real, so $\alpha = 1$, which is not a root of f_n . The second identity gives $\alpha^n = N - n/\alpha$, and therefore $|\alpha|^n \leq N + n/|\alpha| < 2n + 1$. $\square$

Lemma 3 (An inner bound). *If $n \geq 3$, every root α satisfies $|\alpha|^n > 4$. For an upper-half-plane root $\alpha = \rho e^{i\theta}$, with $0 < \theta < \pi$, one also has $\theta > 2\pi/n$.*

Proof. The real and imaginary parts of $N - (n/\rho)e^{-i\theta}$ are positive, so $2\pi k < n\theta < 2\pi k + \pi/2$ for some integer $k \geq 0$. If $k = 0$, every α^j , $1 \leq j \leq n$, has positive imaginary part, contradicting $\sum_{j=1}^n \alpha^j = n$. Hence $\theta > 2\pi/n$.

When $\theta < \pi/2$, squaring the root identity gives

$$\rho^{2n} = (n/\rho - N \cos \theta)^2 + N^2 \sin^2 \theta \geq N^2 \sin^2 \theta.$$

Since $\sin \theta > 2\theta/\pi$ for $0 < \theta < \pi/2$, we obtain

$$\rho^n \geq N \sin \theta > 2N\theta/\pi > 4N/n > 4.$$

When $\theta \geq \pi/2$, the same root identity gives $\rho^n > N$. There are no positive real roots, since all coefficients are positive. A negative real root satisfies $\rho^n = N + n/\rho > N \geq 4$. Conjugation handles the lower half-plane. $\square$

Lemma 4 (The sharper inner bound). *If $n \geq 8$, every root α satisfies*

$$n \ln |\alpha| > 11/6.$$

Proof. In the acute-angle case of Lemma 3, $|\alpha|^n > N \sin(2\pi/n) > 2\pi$. The last inequality follows from $\sin t \geq t - t^3/6$ and $n^2/(n+1) \geq 64/9 > 2\pi^2/3$. The other cases give $|\alpha|^n > N \geq 9 > 2\pi$. Finally $2\pi > e^{11/6}$, so taking logarithms proves the claim. $\square$

2 Factors, coefficients, and the discriminant identity

By Gauss's lemma, a factorization over $\mathbb{Q}$ of a monic integer polynomial can be chosen with monic integer factors. Here *monic* means that the leading coefficient is one. Suppose, then, that $f_n = gh$, where $g, h \in \mathbb{Z}[x]$ are monic and nonconstant. Until a smaller-degree factor is explicitly chosen, either factor may be called g . Write

$$d = \deg g, \quad a = g(0), \quad b = h(0), \quad g_j = [x^j]g.$$

Thus g_j means the coefficient of x^j ; in particular g_1 is the linear coefficient. The equation $ab = n$ will connect these two constant terms to the original parameter. All real roots of the factors are negative. Each conjugate pair contributes $|t - \alpha|^2 > 0$ to evaluation at a real $t \geq 0$. Consequently $g(t), h(t) > 0$ for $t \geq 0$; in particular $a, b, g(1), h(1)$ are positive integers. Lemma 2 gives $a, b > 1$.

Lemma 5 (Coprime constant terms). *Every such factorization satisfies*

$$ab = n, \quad \gcd(a, b) = \gcd(a, g_1) = 1. \quad (2)$$

In particular, $n \geq 6$.

Proof. Compare the constant and linear coefficients: $ab = n$ and $ah'(0) + bg_1 = n - 1$. A common divisor of a, b , or of a, g_1 , would divide both n and $n - 1$. Since $a, b > 1$ are coprime, $ab \geq 2 \cdot 3$. $\square$

The following integer compares the linear coefficient with the coefficient just below the leading one:

$$J = g_1 - ag_{d-1}.$$

Its purpose is to turn the narrow annulus into a small nonzero integer. Later we will show that many primes must divide this same integer.

Lemma 6 (The coefficient difference). *If $\alpha_1, \dots, \alpha_d$ are the roots of g , then*

$$J = a \sum_{i=1}^d (\alpha_i - \alpha_i^{-1}), \quad \gcd(a, J) = 1.$$

In particular $J \neq 0$.

Proof. The root formulas for the two coefficients are $g_{d-1} = -\sum_i \alpha_i$ and $g_1 = -a \sum_i \alpha_i^{-1}$. Also $J \equiv g_1 \pmod{a}$, so Lemma 5 gives $\gcd(a, J) = 1$. Since $a > 1$, this excludes $J = 0$. $\square$

Lemma 7 (The size of the coefficient difference). *Every hypothetical factorization satisfies*

$$0 < |J| < \frac{21}{10} a \ln a. \quad (3)$$

If $n \geq 1000$, the bound improves to $|J| < \frac{201}{100} a \ln a$.

Proof. For a root close to the unit circle, α is close to $\bar{\alpha}^{-1}$: the two numbers point in the same direction, and their distance is $|\alpha| - |\alpha|^{-1}$. Write $\sinh t = (e^t - e^{-t})/2$ for the hyperbolic sine. Conjugation permutes the roots, so Lemma 6 gives

$$|J| = a \left| \sum_i (\alpha_i - \bar{\alpha}_i^{-1}) \right| \leq a \sum_i 2 \sinh(\ln |\alpha_i|).$$

Since $\sinh t/t$ increases and $\sum_i \ln |\alpha_i| = \ln a$, Lemma 2 gives

$$|J| \leq \frac{2 \sinh(\ln(2n+1)/n)}{\ln(2n+1)/n} a \ln a.$$

The quantity $\ln(2n+1)/n$ decreases with n ; it is below $1/2$ for $n \geq 6$ and below $1/100$ for $n \geq 1000$. Here $n \geq 6$ by Lemma 5. The inequality $\sinh t/t \leq (1 - t^2/6)^{-1}$, for $0 < t \leq 1/2$, follows by comparing power series. The two resulting factors are at most $48/23 < 21/10$ and $120000/59999 < 201/100$, respectively. $\square$

Lemma 8 (The derivative at a root). *At every root α of f_n ,*

$$F'(\alpha) = \frac{nN(\alpha - 1)}{\alpha}, \quad f'_n(\alpha) = \frac{nN}{\alpha(\alpha - 1)}.$$

In particular, all roots of f_n are simple.

Proof. By (1),

$$F'(\alpha) = N(\alpha^n - 1) = N(n - n/\alpha) = \frac{nN(\alpha - 1)}{\alpha}.$$

Differentiating $F = (x-1)^2 f_n$ at a root gives $F'(\alpha) = (\alpha-1)^2 f'_n(\alpha)$. Lemma 2 ensures $\alpha \neq 0, 1$, so division is legitimate and the resulting derivative is nonzero. $\square$

We now define the two integer products used in the main comparison. For the distinct roots $\alpha_1, \dots, \alpha_d$ of g ,

$$\text{Disc}(g) = \prod_{i < j} (\alpha_i - \alpha_j)^2, \quad \text{Res}(g, h) = \prod_i h(\alpha_i) = \prod_{g(\alpha)=0} \prod_{h(\beta)=0} (\alpha - \beta).$$

The *discriminant* measures differences within one factor; the *resultant* uses differences between the two factors. Both products are nonzero because Lemma 8 rules out repeated roots of f_n . Their absolute values are products of ordinary distances.

Both are integers. To see why, use the elementary theorem that a symmetric polynomial with integer coefficients is an integer polynomial in the elementary symmetric functions. Viète's

formulas identify those functions with the coefficients of the monic polynomial. Apply this first to the roots of g , and, for the resultant, also to those of h . Separating root pairs also gives the useful identity

$$\text{Disc}(gh) = \text{Disc}(g) \text{Disc}(h) \text{Res}(g, h)^2.$$

For a monic quadratic, the discriminant is the familiar linear coefficient squared minus four times the constant coefficient.

Lemma 9 (The discriminant–resultant identity). *Every nontrivial monic integer factorization satisfies*

$$|\text{Disc}(g)| |\text{Res}(g, h)| a g(1) = (nN)^d. \quad (4)$$

Consequently, every prime divisor of $\text{Res}(g, h)$ divides nN .

Proof. Write $g(x) = \prod_{i=1}^d (x - \alpha_i)$. Multiplying the identity in Lemma 8 gives

$$\left(\prod_i f'_n(\alpha_i) \right) \left(\prod_i \alpha_i \right) \left(\prod_i (\alpha_i - 1) \right) = (nN)^d.$$

Evaluation of g at 0 and 1 shows that

$$\prod_i \alpha_i = (-1)^d a, \quad \prod_i (\alpha_i - 1) = (-1)^d g(1).$$

These two signs cancel. Since $f_n = gh$ and $g(\alpha_i) = 0$,

$$\prod_i f'_n(\alpha_i) = \left(\prod_i g'(\alpha_i) \right) \left(\prod_i h(\alpha_i) \right).$$

For monic g , the second product is $\text{Res}(g, h)$. In the first, each unordered root pair contributes $(\alpha_i - \alpha_j)(\alpha_j - \alpha_i) = -(\alpha_i - \alpha_j)^2$. Thus

$$\prod_i g'(\alpha_i) = (-1)^{d(d-1)/2} \text{Disc}(g).$$

Substitution gives the signed identity

$$(-1)^{d(d-1)/2} \text{Disc}(g) \text{Res}(g, h) a g(1) = (nN)^d.$$

Take absolute values and use $a, g(1) > 0$ to obtain (4). All its factors are nonzero integers, proving the assertion about prime divisors. $\square$

3 Local root packets and resultant bounds

The resultant is an integer, so its size is determined by its prime factorization. We now bound the exponent of each possible prime divisor. This section introduces the local-field tools used below.

For a nonzero rational number, v_p is the exponent of p in its numerator minus the exponent in its denominator. For example, $v_p(p) = 1$ and $v_p(1/p) = -1$; put $v_p(0) = +\infty$. The essential rules are

$$v_p(xy) = v_p(x) + v_p(y), \quad v_p(x + y) \geq \min\{v_p(x), v_p(y)\},$$

with equality in the second rule when the two valuations differ. Thus a sum cannot be zero if exactly one term has the smallest valuation. This is the divisibility analogue of a term being too large to cancel.

The field $\mathbb{Q}_p$ completes rational arithmetic for the notion of closeness measured by divisibility by high powers of p . Its valuation extends uniquely to algebraic extensions, with the same rules.

We write v for this extension. It may take fractional values at algebraic roots, even though v_p on rational numbers is integer-valued.

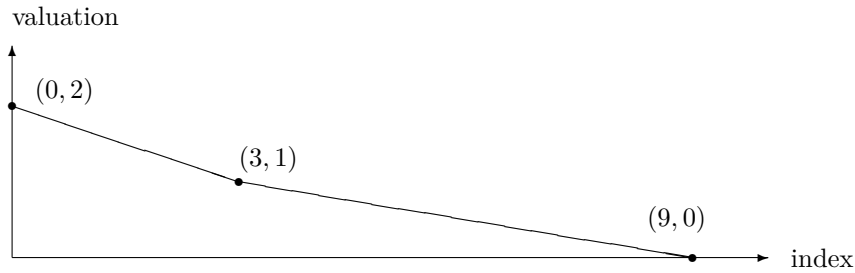
An element is called *integral* if its valuation is nonnegative. A *unit* is an integral element whose inverse is also integral, equivalently an element of valuation zero. Integral elements can be reduced to a residue field of characteristic p : two have the same residue exactly when their difference has positive valuation. For ordinary integers, this agrees with reduction modulo p . In particular, different residues have a difference of valuation zero.

We use two standard local-field tools in the following forms.¹

Unramified extensions and Hensel lifting. If $p \nmid m$, there is a finite extension $E/\mathbb{Q}_p$ containing all m th roots of unity, with $v(E^\times) = \mathbb{Z}$ and $v(p) = 1$. This is an unramified extension. It is obtained from a finite residue field containing these roots; Hensel's theorem lifts simple roots of the reduction to roots of the polynomial over E .

Newton polygon theorem. For a monic polynomial over E with nonzero constant term, plot each coefficient's index against its valuation, omitting zero coefficients. Take the lower convex boundary of these points. Each segment's horizontal length counts roots, with multiplicity; their valuation is the negative of its slope. The roots belonging to one segment form a monic factor over E . We call such a factor a *packet*.

For example, the three points below give a segment of length three and slope $-1/3$, followed by one of length six and slope $-1/6$. They describe three roots of valuation $1/3$ and six of valuation $1/6$.



Applying the theorem to a translated polynomial, with $y = x - \zeta$, describes the valuations $v(\alpha - \zeta)$ of roots near ζ . Consecutive collinear points belong to one segment; this will matter when $p = 2$.

For a prime $p \mid nN$, write the member of $\{n, N\}$ divisible by p as $p^r m$, where $p \nmid m$. Use the field E just described. The condition $p \nmid m$ ensures that the roots of $x^m - 1$ are distinct in the residue field.

Lemma 10 (Residue classes and the root reducing to zero). *Every root of f_n is integral and reduces either to zero or to an m th root of unity. For an m th root of unity ζ , its residue class contains p^r roots if $\zeta \neq 1$, and $p^r - 2$ if $\zeta = 1$. The zero class contains exactly one root, of valuation r , if $p \mid n$; otherwise it contains none.*

Proof. Monicity gives integrality: a root of negative valuation would make the leading term the unique term of smallest valuation, preventing cancellation. Modulo p , the polynomial F is $(x^m - 1)^{p^r}$ if $p \mid N$, and $x(x^m - 1)^{p^r}$ if $p \nmid n$. The roots of $x^m - 1$ are distinct because $p \nmid m$. Removing $(x - 1)^2$ gives the stated counts for f_n . When $p \mid n$, the Newton polygon segment of F from $(0, r)$ to the unit linear coefficient at $(1, 0)$ gives valuation r for the single residue-zero root. $\square$

The next observation explains why fractional root valuations can force irreducibility over E : selecting too few roots would produce a coefficient whose valuation is not an integer.

¹Proofs are in J. S. Milne, *Algebraic Number Theory*, version 3.01, Theorem 7.38, Corollary 7.40, Propositions 7.44 and 7.50, and Corollary 7.51. Milne orders coefficients from the leading term; our horizontal coordinate is the power of x , so the Newton slopes have opposite signs.

Lemma 11 (A denominator criterion). *Let $P \in E[x]$ be monic, $\zeta \in E$, and $e \geq 1$ an integer. If every root α of P satisfies $v(\alpha - \zeta) = 1/e$, then every monic divisor of P in $E[x]$ has degree divisible by e . In particular, P is irreducible if $\deg P = e$.*

Proof. For a monic divisor A , multiplication over its roots gives $v(A(\zeta)) = \deg(A)/e$. This belongs to the value group $\mathbb{Z}$ of E , so $e \mid \deg A$. $\square$

For the unit roots, the common coefficient calculation is

$$\begin{aligned} v_p\binom{N}{j} &= r - v_p(j), & p \mid N, \quad 1 \leq j \leq p^r, \\ v_p\binom{n+1}{j} &= r - v_p(j(j-1)), & p \mid n, \quad 2 \leq j \leq p^r + 1. \end{aligned}$$

To verify them, use

$$\binom{N}{j} = \frac{N}{j} \binom{N-1}{j-1}, \quad \binom{n+1}{j} = \frac{n(n+1)}{j(j-1)} \binom{n-1}{j-2}.$$

The last binomial coefficient in each relevant formula is a unit: $\binom{p^r m - 1}{k} \equiv (-1)^k \pmod{p}$ for $k < p^r$. Indeed, each factor $(p^r m - i)/i$ in its product formula is -1 modulo p , since $v_p(i) < r$. At residue one we remove the double root by using

$$f_n(1+y) = \sum_{k=0}^{n-1} \binom{n+1}{k+2} y^k. \quad (5)$$

For $\zeta \neq 1$, the constant term of $F(\zeta + y)$ is $N(1 - \zeta)$ if $p \mid N$, and $n(1 - \zeta)$ if $p \mid n$. Its valuation is therefore r . The linear coefficient is

$$F'(\zeta) = \begin{cases} 0, & p \mid n, \\ N(\zeta^{-1} - 1), & p \mid N, \end{cases}$$

and has valuation r in the second case. Together with the higher coefficient valuations above, this determines the following three simple polygon shapes.

Lemma 12 (Packets at odd primes). *Suppose p is odd and ζ is an m th root of unity. The roots reducing to ζ form irreducible packets. The first has degree $e_1 = p$ if $\zeta \neq 1$, and $e_1 = p - 2$ if $\zeta = 1$. The subsequent degrees are*

$$e_j = p^{j-1}(p-1) \quad (2 \leq j \leq r).$$

Every root of packet j satisfies $v(\alpha - \zeta) = 1/e_j$.

Proof. For $\zeta \neq 1$, the coefficient valuations above give the negative-slope vertices of $F(\zeta + y)$ as

$$(0, r), (p, r-1), \dots, (p^r, 0).$$

For $\zeta = 1$, formula (5) gives instead

$$(0, r), (p-2, r-1), \dots, (p^r-2, 0).$$

The horizontal lengths are the stated e_j , and each segment drops by one, so its slope is $-1/e_j$. The Newton polygon theorem gives the packets; Lemma 11 makes each irreducible. $\square$

Lemma 13 (Binary packets away from one). *Suppose $p = 2$ and $\zeta \neq 1$ is an m th root of unity. If $r = 1$, its residue class consists of one irreducible quadratic packet, with displacement valuation $1/2$. If $r \geq 2$, it consists of a quartic packet with displacement valuation $1/2$, followed by irreducible packets of degrees 2^{j-1} and displacement valuations 2^{1-j} , for $3 \leq j \leq r$. Every factor of the quartic in $E[x]$ has even degree.*

Proof. The negative-slope part of the Newton polygon of $F(\zeta + y)$ is the lower hull of

$$(0, r), (2, r-1), \dots, (2^r, 0).$$

For $r = 1$, its only segment has length two and slope $-1/2$. For $r \geq 2$, the first three points are collinear, giving length four and slope $-1/2$; each later segment has length 2^{j-1} and slope -2^{1-j} . Apply the Newton polygon theorem and Lemma 11. The latter gives only evenness of factor degrees for the quartic, whose degree is four rather than two. $\square$

Lemma 14 (Binary packets at one). *Suppose $p = 2$. The residue-one class is empty if $r = 1$. If $r \geq 2$, it consists of irreducible packets of degrees 2^{j-1} and displacement valuations 2^{1-j} , for $2 \leq j \leq r$.*

Proof. For $r = 1$, use the count in Lemma 10. For $r \geq 2$, the binomial valuations in (5) give the negative-slope vertices

$$(0, r-1), (2, r-2), \dots, (2^r-2, 0).$$

Their lengths and slopes give the stated packets, which are irreducible by Lemma 11. $\square$

In each residue class, these packet degrees sum to the count in Lemma 10, so the lists account for all $n-1$ roots.

Lemma 15 (The local resultant bound). *Define $C_p(0) = C_p(1) = 0$ and $C_p(r) = p^{r-1} + C_p(r-2)$ for $r \geq 2$. Then*

$$v_p \text{Res}(g, h) \leq mC_p(r) - 2\lfloor r/2 \rfloor. \quad (6)$$

Proof. Different residue classes have unit root differences and contribute zero. This includes the residue-zero root, regardless of its allocation. If $r = 1$, Lemmas 12, 13 and 14 give at most one irreducible packet in each unit residue class, so the resultant valuation is zero.

For odd p and $r \geq 2$, a rational factor selects whole packets from Lemma 12: g, h are also polynomials over E , where these packets are irreducible. Within one residue class, packets $i < j$ have different displacement valuations. The valuation rule gives $v(\alpha - \beta) = 1/e_j$ for a root from each packet. There are $e_i e_j$ such pairs, so their total contribution is e_i when the packets go to different factors.

Represent each packet by a vertex, colored according to its assigned factor. Join vertices $i < j$ with an edge of weight e_i . Only edges between different colors contribute. A *cut* is just such a division into two colors, so bounding the resultant has become the elementary problem of maximizing the sum of crossing edge weights.

A maximum cut can place the last two vertices on opposite sides: if they agree, flipping the penultimate vertex gains e_{r-1} and loses at most $\sum_{i < r-1} e_i \leq e_{r-1}$. With those two vertices opposite, each earlier vertex has exactly one crossing edge to them, regardless of its color. Including the edge between the last two vertices, deleting the pair therefore contributes $\sum_{i < r} e_i = p^{r-1}$, giving the recurrence for $C_p(r)$. At residue one the contribution is $p^{r-1} - 2$, so its bound is $C_p(r) - 2\lfloor r/2 \rfloor$.

At $p = 2$, a quartic packet may split into two quadratics, with one assigned to each factor. We must also bound that internal contribution. For the binary quartic P at a nontrivial residue root, described in Lemma 13, Lemma 8 gives $v(F'(\alpha)) = r$ at its roots. Each of the $r-2$ higher packets contributes one to this valuation; the other residue classes contribute zero. Therefore $v(P'(\alpha)) = 2$, and $v(\text{Disc } P) = 8$. If P is split between g and h , both parts are quadratic. After translation their constant coefficients have valuation one and their linear coefficients have valuation at least one. Each quadratic discriminant has valuation at least two: its linear coefficient squared has valuation at least two, and four times its constant coefficient has valuation three. The discriminant product identity then makes the internal resultant valuation at most $(8-2-2)/2 = 2$.

Two formal vertices of weight two, with an internal edge of weight two, therefore bound every allocation of this quartic. Color them alike for a whole allocation, and differently for a quadratic split. This model allows every actual allocation without requiring a split to exist. Their edges to later packets have exactly the stated weights. The cut calculation applies to $2, 2, 4, 8, \dots$; for the packets of Lemma 14, the initial formal weight is zero. Summing over the m residue classes proves (6). $\square$

Lemma 16 (Prime divisibility of the coefficient difference). *For every hypothetical factorization,*

$$\text{rad}(bN) \mid J. \quad (7)$$

Proof. If $p \mid bN$, then $p \nmid a$, so all roots of g are units. Reduce in the residue field of a valued splitting extension. By Lemmas 12 and 13, the selected degree at each nontrivial residue root is divisible by p , including a possible quadratic part of the binary quartic. Its contribution to $\sum_i (\alpha_i - \alpha_i^{-1})$ therefore vanishes in the residue field. For clarity, all roots in a packet reduce to the same ζ , so the sum of $\alpha - \alpha^{-1}$ over those roots reduces to the packet's degree times $(\zeta - \zeta^{-1})$; that is zero when the degree is divisible by p . Each summand at residue one also vanishes. Lemma 6 now gives $v_p(J) > 0$; since J is an integer, $p \mid J$. Taking the product over the distinct primes proves the claim. $\square$

To turn bounds on prime exponents into a size bound, use $\ln |\text{Res}(g, h)| = \sum_p v_p(\text{Res}(g, h)) \ln p$. For these global estimates write

$$R = \ln |\text{Res}(g, h)|, \quad Q = \prod_{p^2 \mid nN} p.$$

Thus Q records the distinct primes whose squares divide nN . The case $r = 1$ of Lemma 15 explains why only these primes can contribute to the resultant.

Lemma 17 (Two upper bounds for the resultant). *For every hypothetical factorization,*

$$R \leq N \sum_{p \mid Q} \frac{p \ln p}{p^2 - 1}. \quad (8)$$

For $n \geq 15$ one also has

$$R < \left(\frac{N}{16} - 1 \right) \ln(nN) + \frac{13N}{20}. \quad (9)$$

Proof. By Lemma 9, only primes dividing nN occur. The recurrence in Lemma 15 gives

$$\frac{C_p(r)}{p^r} = \sum_{j=1}^{\lfloor r/2 \rfloor} \frac{1}{p^{2j-1}} < \frac{p}{p^2 - 1}.$$

Since $C_p(1) = 0$, this proves (8).

Retain the negative term in (6), and use $p^r m \leq N$ in its positive term. Subtracting $1/8$ from each summand in the displayed series leaves positive excess only in the first layers for $p = 2, 3, 5, 7$. Hence

$$\begin{aligned} R &\leq (N/8 - 2) \sum_{p \mid nN} \lfloor v_p(nN)/2 \rfloor \ln p \\ &\quad + N \left(\frac{3}{8} \ln 2 + \frac{5}{24} \ln 3 + \frac{3}{40} \ln 5 + \frac{1}{56} \ln 7 \right). \end{aligned}$$

Since $N \geq 16$, we can use $\sum_{p \mid nN} \lfloor v_p(nN)/2 \rfloor \ln p \leq \frac{1}{2} \ln(nN)$. The bounds $\ln 2 < 0.7$, $\ln 3 < 1.1$, $\ln 5 < 13/8$ and $\ln 7 < 2$ put the four-logarithm sum below $4363/6720 < 13/20$. These two estimates give (9). $\square$

4 The main comparison for $n \geq 182$

We now return to ordinary complex absolute values. An upper bound on the discriminant, together with the exact identity (4), forces the resultant to be large. The preceding section bounds that same resultant from above. We will make the contradiction explicit using logarithms, which turn products into sums.

Choose g of smaller degree and put

$$L = \ln(2n + 1), \quad A = \ln a.$$

Thus L/n bounds the logarithmic root moduli, while A is their sum for the factor g . The annulus gives $0 < A/L < d/n < 1/2$. We first establish estimates for all such factorizations. The assumption $a, b \geq 13$ will enter only in Proposition 24.

Lemma 18 (The complex discriminant estimate). *If the logarithms of the root moduli of g lie in an interval of width at most W/n , where $W > 0$, then*

$$\ln |\text{Disc}(g)| \leq (d-1)A + \frac{Wd^2}{2n} - d \ln(W/n). \quad (10)$$

Proof. The roots α_i are nonzero and distinct, so the logarithms of their moduli and pairwise distances below are defined. Put $u_i = \alpha_i/|\alpha_i|$ and $t = e^{-W/n}$. Thus $|u_i| = 1$ and $0 < t < 1$. For $i \neq j$,

$$\frac{|\alpha_i - \alpha_j|^2}{|\alpha_i| |\alpha_j|} = \frac{|\alpha_i|}{|\alpha_j|} + \frac{|\alpha_j|}{|\alpha_i|} - 2 \operatorname{Re}(u_i \overline{u_j}).$$

The ratio of the two moduli lies between t and t^{-1} , by the assumption on their logarithms. Its sum with its reciprocal is at most $t + t^{-1}$, giving

$$\frac{|\alpha_i - \alpha_j|^2}{|\alpha_i| |\alpha_j|} \leq t^{-1} |1 - t u_i \overline{u_j}|^2.$$

We use the real logarithmic series

$$\ln |1 - tz| = - \sum_{k \geq 1} \frac{t^k}{k} \operatorname{Re}(z^k) \quad (|z| = 1, 0 < t < 1).$$

It follows by taking real parts of the geometric series and integrating in t . Absolute convergence allows us to sum over the finitely many root pairs. The identity $\sum_{i,j} u_i^k \overline{u_j^k} = |\sum_i u_i^k|^2$ gives

$$\sum_{i \neq j} \ln |1 - t u_i \overline{u_j}| = - \sum_{k \geq 1} \frac{t^k}{k} \left| \sum_i u_i^k \right|^2 - d \ln(1 - t).$$

The term $-d \ln(1 - t)$ removes the d diagonal terms $i = j$. Sum the pairwise inequality and discard the nonpositive series to get

$$\ln |\text{Disc}(g)| \leq (d-1)A + \frac{Wd(d-1)}{2n} - d \ln(1 - t).$$

Here each root modulus occurs in $d-1$ pairs, and there are $d(d-1)/2$ unordered pairs. Using $1 - e^{-W/n} \geq (W/n)e^{-W/(2n)}$ then gives (10): the two quadratic contributions combine as $d(d-1) + d = d^2$. $\square$

For such a width, abbreviate the expression that will occur in our lower bound for the resultant by

$$G_W = \frac{d}{n} (\ln n - A + \ln W) - \frac{Wd^2}{2n^2}.$$

Lemma 2 permits $W = L$; Lemma 4 also permits $W = L - 11/6$ for $n \geq 8$.

Lemma 19 (Combining the two root products). *For every admissible width W/n ,*

$$G_W \leq \frac{R + \ln g(1)}{n} - \frac{d}{n} \ln(N/n). \quad (11)$$

Proof. Taking logarithms of (4) gives $\ln |\text{Disc}(g)| + R + A + \ln g(1) = d \ln(nN)$. Insert (10), rearrange, and divide by n . Explicitly, this gives

$$\frac{R + \ln g(1)}{n} \geq \frac{d}{n} (\ln N - A + \ln W) - \frac{Wd^2}{2n^2} = G_W + \frac{d}{n} \ln(N/n).$$

Thus G_W packages a necessary lower estimate; no new assumption has been introduced. $\square$

Lemma 20 (The uniform upper bound). *If $n \geq 182$, then for every admissible width,*

$$G_W < \frac{1}{8} \ln n + 33/50. \quad (12)$$

Proof. Since $g(1)h(1) = nN/2$ and both factors are positive integers, $\ln g(1) \leq \ln(nN) - \ln 2$. In (11), this cancels the $-\ln(nN)$ term of (9). Use $\ln(nN) < 2 \ln n + 1/n$. The error above $\frac{1}{8} \ln n + 13/20$ is at most

$$\frac{\frac{1}{8} \ln n + 13/20 + 1/16 + 1/(16n)}{n} < 1/100 \quad (n \geq 182),$$

proving the stated bound. For this last numerical inequality, all the terms $(\ln n)/n$, $1/n$ and $1/n^2$ decrease for $n \geq 182$. It therefore suffices to use $n = 182$ and $\ln 182 < 6$. $\square$

For the next bound write

$$S(A) = \frac{3}{20}A + \frac{1}{10} \ln(2.01A) + \frac{97}{100}.$$

Lemma 21 (The upper bound from prime support). *If $n \geq 1000$, then*

$$G_L < S(A) + 1/50. \quad (13)$$

Proof. If $p \mid Q$ and $p \mid a$, then $p^2 \mid a$, since $\gcd(a, bN) = 1$. The product of these distinct primes is consequently at most $\sqrt{a}$. All other primes of Q divide bN . By Lemmas 16 and 7,

$$Q \leq \sqrt{a} \text{rad}(bN) \leq \sqrt{a} |J| < 2.01a^{3/2}A.$$

For $p \geq 11$, $p/(p^2 - 1) \leq 1/10$, while

$$\sum_{p=2,3,5,7} \left(\frac{p}{p^2 - 1} - \frac{1}{10} \right) \ln p < 97/100.$$

Separating the four small primes from the rest gives

$$\sum_{p \mid Q} \frac{p \ln p}{p^2 - 1} < \frac{\ln Q}{10} + \frac{97}{100} < S(A).$$

Combine (8) with (11). The extra terms from $N/n = 1 + 1/n$ and $\ln g(1)/n$ give

$$\begin{aligned} G_L &< S(A) + \frac{\frac{1}{10} \ln Q + \frac{97}{100} + \ln g(1)}{n} \\ &< S(A) + \frac{2.1 \ln n + 1}{n} < S(A) + \frac{1}{50}. \end{aligned}$$

For the second line use $Q \leq \sqrt{nN}$, $\ln g(1) \leq \ln(nN) - \ln 2$, and $\ln(nN) < 2 \ln n + 1/n$. Again the error decreases with n , so $n = 1000$ and $\ln 1000 < 7$ are sufficient to check the constant. $\square$

Only elementary calculus remains in the main comparison. We repeatedly use two facts: an increasing function is bounded below by its left endpoint value, and a concave function lies above the chord joining its endpoints. In particular, a concave function is bounded below by the smaller endpoint value. These facts let a few exact endpoint estimates cover entire intervals.

Lemma 22 (Comparison for large parameters). *If $n \geq 1000$ and $51/20 \leq A < dL/n < L/2$, the two bounds (12) with $W = L$ and (13) are incompatible.*

Proof. Here $\ln n > 6.9$, $0 < L - \ln n < 0.7$ and $\ln L > 2$. Regard G_L as a quadratic in the real ratio d/n , holding n, A, L fixed. Its derivative is $\ln n - A + \ln L - L(d/n)$, which is greater than $\ln L - (L - \ln n) > 0$. Thus replacing d/n by its smaller lower bound A/L gives

$$G_L > H(A) := \frac{A(\ln n + \ln L - 3A/2)}{L}.$$

The first two rows of the following table will contradict the upper bound $S(A) + 0.02$; the last will contradict $\frac{1}{8} \ln n + 0.66$. Together the rows cover all possible A :

range of A	lower bound
$[51/20, 3]$	$H(A) - S(A) > 1/10$
$[3, \frac{1}{4} \ln n]$	$H(A) - S(A) > 51/200$
$[\max(3, \frac{1}{4} \ln n), L/2]$	$H(A) - \frac{1}{8} \ln n > 59/80$.

An interval with reversed endpoints is ignored. For the first row, $(\ln n + 2 - 1.5A)/(\ln n + 0.7)$ increases with $\ln n$, so

$$H(A) > \frac{A(8.9 - 1.5A)}{7.6}.$$

Subtract $S(A)$ from the right-hand side. The resulting function has second derivative $-3/7.6 + 1/(10A^2) < 0$, so it is concave. Its endpoint values exceed 0.18 and 0.1, using $\ln(2.55 \cdot 2.01) < 1.7$ and $\ln 6.03 < 2$. For the second row, $H(A) > 5A/8$, and $19A/40 - \ln(2.01A)/10 - 0.97$ has derivative $19/40 - 1/(10A) > 0$ for $A \geq 3$ and exceeds 0.255 at three.

For the last row, $H''(A) = -3/L < 0$, so only its two endpoints matter. At $A = L/2$,

$$H(A) - \frac{1}{8} \ln n = \frac{1}{2} \ln L - \frac{3}{8}(L - \ln n) > 59/80.$$

The other endpoint is three when $\ln n \leq 12$, and $\frac{1}{4} \ln n$ when $\ln n \geq 12$. For $6.9 \leq \ln n \leq 12$, a lower bound is $3(\ln n - 2.5)/(\ln n + 0.7) - \frac{1}{8} \ln n$. This is concave in $\ln n$, with endpoint values $1329/1520$ and $189/254$, both greater than $59/80$. For $\ln n \geq 12$, the lower bound is $\frac{\ln n}{\ln n + 0.7}(\frac{1}{32} \ln n + 33/80) \geq 189/254$. Both factors increase with $\ln n$, so evaluating at twelve suffices. The first two rows contradict (13); the last contradicts (12). $\square$

Lemma 23 (Comparison in the intermediate range). *If $182 \leq n < 1000$, $51/20 \leq A < dL/n < L/2$, and $W = L - 11/6$, then*

$$G_W > \frac{1}{8} \ln n + 2/3.$$

Proof. Here $5.2 < \ln n < 7$, $0.69 < L - \ln n < 0.7$ and $1.4 < \ln W < 1.8$. The derivative of G_W with respect to the real ratio d/n is $\ln n - A + \ln W - W(d/n)$. Since $W \leq L$, it exceeds $\ln W - (L - \ln n) > 0.7$. Thus G_W increases with d/n , giving

$$G_W > H_W(A) := \frac{A(\ln n + \ln W)}{L} - \frac{A^2(L + W/2)}{L^2}.$$

For $51/20 \leq A \leq L/2$, we have $H_W(A) \geq H_W(L/2)$. Indeed,

$$\begin{aligned} H_W(A) - H_W(L/2) &= \frac{A - L/2}{L} \\ &\quad \times [\ln n + \ln W - (A + L/2)(1 + W/(2L))]. \end{aligned}$$

The bracket is negative, because

$$\begin{aligned} & 2L [\ln n + \ln W - (A + L/2)(1 + W/(2L))] \\ & \leq L \left(2 \ln n - \frac{3}{2}L - \frac{47}{15} \right) + \frac{187}{40} \leq -\frac{27869}{60000} < 0. \end{aligned}$$

The first bound uses $A \geq 51/20$, $\ln W < 1.8$ and $W = L - 11/6$. For the second, the polynomial expression decreases with L . Substituting $L = \ln n + 0.69$ gives an increasing function of $\ln n$ on $[5.2, 7]$, whose value at seven is $-27869/60000$. The prefactor $(A - L/2)/L$ is nonpositive, proving the desired comparison with $H_W(L/2)$. Consequently

$$\begin{aligned} G_W & > H_W(L/2) \\ & = \frac{1}{8} \ln n + \frac{1}{2} \ln W - \frac{3}{8}(L - \ln n) + \frac{11}{48} \\ & > \frac{1}{8} \ln n + \frac{2}{3}. \end{aligned}$$

All terminating decimals in these comparisons are exact rational constants; the logarithmic bounds follow from the exponential series. $\square$

Proposition 24 (The main obstruction). *There is no nontrivial monic integer factorization $f_n = gh$ with $g(0), h(0) \geq 13$.*

Proof. If $a, b \geq 13$, Lemma 5 gives $n = ab \geq 13 \cdot 14 = 182$. Choose g of smaller degree. Then $A \geq \ln 13 > 51/20$ and $A < dL/n < L/2$. For $n \geq 1000$, Lemmas 20, 21 and 22 contradict one another. For $182 \leq n < 1000$, Lemma 23 contradicts Lemma 20, since $2/3 > 33/50$. $\square$

5 Excluding small constant terms and completing the proof

It remains to rule out factors of constant term $2, \dots, 12$. The arguments in this section apply to every $n \geq 2$, so they also settle all $n < 182$. We again allow either factor to be called g . We will combine the small size of J with the requirement $\text{rad}(bN) \mid J$. The two lemmas after the parity observation provide the elementary congruences needed for the few remaining possibilities.

Lemma 25 (Parity of a factor degree). *Every hypothetical factorization satisfies*

$$d \equiv \mathbf{1}_{2 \mid a} \pmod{2}.$$

Proof. At $p = 2$, Lemmas 13 and 14 show that every selected unit packet has even degree. By Lemma 10, the residue-zero root exists only when $2 \mid n$; it belongs to g exactly when $2 \mid a$, by taking valuations of the product of its roots. This proves the congruence. $\square$

Lemma 26 (A product identity and lower bound). *Every hypothetical factorization satisfies*

$$a^N = N^d g(n/N) > (\sqrt{nN})^d. \quad (14)$$

Proof. The integer nd is even. This is automatic if n is even; if n is odd, then N is even and $F(x) > 0$ for $x < 0$, so g has no real roots and d is even. Multiplying $\alpha^N = N(\alpha - n/N)$ over the roots of g now gives $a^N = N^d g(n/N)$, with the signs cancelling because nd is even. For $0 < t < 1$ and $|z| > 1$,

$$|z - t|^2 - t|z - 1|^2 = (1 - t)(|z|^2 - t) > 0.$$

Apply this with $t = n/N$ and multiply over the roots to obtain

$$g(n/N) > (n/N)^{d/2} g(1) \geq (n/N)^{d/2}.$$

Multiplication by N^d gives the stated lower bound. $\square$

Lemma 27 (Two congruences for the constant term). *Every hypothetical factorization satisfies*

$$a^N \equiv (-1)^d \pmod{N}, \quad a^n \equiv 1 \pmod{b}. \quad (15)$$

Proof. Expand $N^d g(n/N)$ in (14). Modulo N only its leading term n^d survives, giving the first congruence.

For the second congruence we use a matrix version of polynomial long division. Let C send the coefficient vector of a polynomial of degree below d to the coefficient vector of the remainder of its product with x , after division by g . Relative to $1, x, \dots, x^{d-1}$, its first $d-1$ columns shift one basis vector to the next, and its last column consists of the negated coefficients $-g_j$, $0 \leq j < d$. Thus C is an integer matrix with

$$g(C) = 0, \quad \det C = (-1)^d a.$$

The first identity says that multiplication by $g(x)$ always leaves zero remainder. Since g divides F , it also gives $F(C) = 0$. Modulo b , where $n \equiv 0$ and $N \equiv 1$, this becomes $C^{n+1} - C = 0$. The determinant is invertible modulo b because $\gcd(a, b) = 1$; the adjugate formula therefore gives an inverse matrix. Multiplying by it yields $C^n = I$. Taking determinants and using even nd now gives $a^n \equiv 1 \pmod{b}$. $\square$

First we show that $N = n+1$ cannot be a prime power in a hypothetical factorization. This will make N supply at least two distinct primes to J , and b a third. For $N = p^r$, the *cyclotomic polynomial*

$$\Phi_N(x) = \frac{x^N - 1}{x^{N/p} - 1} = 1 + x^{N/p} + \dots + x^{(p-1)N/p}$$

has as its roots the primitive N th roots of unity: those whose multiplicative order is exactly N . Its degree is $\varphi(N) = N - N/p$, and $\Phi_N(1) = p$. Reduction modulo p gives $\Phi_N(x) \equiv (x-1)^{\varphi(N)}$, since powers of p satisfy $(x-1)^{p^j} = x^{p^j} - 1$ in characteristic p . These explicit formulas provide the cyclotomic facts used below.

Lemma 28 (A cyclotomic resultant). *Suppose $N = p^r$ is a prime power and g is the smaller-degree factor of a hypothetical factorization. Then*

$$|\text{Res}(g, \Phi_N)| = p^d,$$

where Φ_N is the N th cyclotomic polynomial.

Proof. We compute the power of p in this resultant and then show that no other prime divides it. Use the valuation at p . Since g is the smaller factor, $0 < d < N/2 \leq \varphi(N)$. The reduction of f_n is $(x-1)^{N-2}$, so the monic reduction of g is $(x-1)^d$. Every root α therefore has $v(\alpha-1) > 0$.

All nonleading coefficients of $g(1+y)$ are divisible by p . At $y = \alpha - 1$, every nonleading term has valuation at least one. The leading term cannot be the unique term of smaller valuation; hence $d v(\alpha - 1) \geq 1$. Similarly, $\Phi_N(x) \equiv (x-1)^{\varphi(N)}$ modulo p . Thus $\Phi_N(1+y)$ is monic, all its nonleading coefficients are divisible by p , and every primitive root reduces to one. Its constant coefficient is $\Phi_N(1) = p$. For a primitive root ζ , we therefore have $v(\zeta - 1) > 0$. All terms of intermediate degree then have valuation greater than one. The constant and leading terms must have equal valuation, so

$$v(\alpha - 1) \geq \frac{1}{d} > \frac{1}{\varphi(N)} = v(\zeta - 1).$$

The unequal-valuation rule gives $v(\alpha - \zeta) = 1/\varphi(N)$. Multiplying over the $\varphi(N)$ primitive roots shows that $v(\Phi_N(\alpha)) = 1$, and then over the d roots of g that $v_p(\text{Res}(g, \Phi_N)) = d$.

For the prime divisors, evaluate (1) at a primitive root: $f_n(\zeta) = N/(1-\zeta)$. Also $\prod_{\Phi_N(\zeta)=0} (1-\zeta) = \Phi_N(1) = p$. Consequently

$$|\text{Res}(f_n, \Phi_N)| = \left| \prod_{\Phi_N(\zeta)=0} f_n(\zeta) \right| = N^{\varphi(N)} / p.$$

The root-product definition gives $\text{Res}(f_n, \Phi_N) = \text{Res}(g, \Phi_N) \text{Res}(h, \Phi_N)$. Both factors are integers, so the resultant for g divides this pure power of p . Its already computed valuation proves the claim. $\square$

Lemma 29 (Prime-power successors). *If $n \geq 2$ and $n+1$ is a prime power, then f_n is irreducible.*

Proof. Suppose $N = p^r$ and choose g of smaller degree. For odd p and $r \geq 3$, Lemma 3 gives

$$|\Phi_N(\alpha)| \geq \frac{|\alpha|^N - 1}{|\alpha|^{N/p} + 1} > \frac{3}{8} |\alpha|^{\varphi(N)}.$$

For the last step, factor out $|\alpha|^{\varphi(N)}$. The remaining numerator $1 - |\alpha|^{-N}$ is greater than $3/4$, and the denominator $1 + |\alpha|^{-N/p}$ is less than two. Multiply over the roots of g and use Lemmas 28 and 26. This yields $p > \frac{3}{8}(nN)^{(p-1)/(2p)}$, whereas $\sqrt{nN} > p^3 - 1 > \frac{8}{9}p^3$ gives $(nN)^{(p-1)/(2p)} > \frac{8}{9}p^2 \geq \frac{8}{3}p$, a contradiction.

For $p = 2$, use $\Phi_N(x) = x^{N/2} + 1$ instead. Since $|\alpha|^{N/2} > 2$, its absolute value is greater than $|\alpha|^{N/2}/2$. The same lemmas give $2^d > a^{N/2}/2^d > ((nN)^{1/4}/2)^d$, hence $nN < 256$. This excludes $N \geq 32$. For $N = 4, 8$ the constant n is prime. For $N = 16$, the constants would be 3, 5, contradicting $5^{15} \equiv 1 \pmod{3}$ in (15).

If $N = p^2$ with odd p , one factor has even degree and the other odd degree, since their degrees sum to $p^2 - 2$. Fermat's congruence $a^{p^2} \equiv a \pmod{p}$ and (15) show that their constants are respectively 1 and -1 modulo p , so they are at least $p + 1$ and $p - 1$. Their product $p^2 - 1$ forces equality, contradicting coprimality. Finally, if $N = p$ is odd prime, $f_n(x + 1)$ is Eisenstein at p . Indeed, every nonleading coefficient in (5) is divisible by p , and the constant $p(p - 1)/2$ is not divisible by p^2 . Eisenstein's criterion says that these conditions force irreducibility over $\mathbb{Q}$. The value $N = 2$ is excluded by $n \geq 2$. $\square$

Lemma 30 (Small constants other than eleven). *In any hypothetical factorization, a constant term below 13 must equal 11.*

Proof. The integers a, b, N are pairwise coprime. By Lemma 29, N has at least two distinct prime divisors. Lemmas 16 and 7 give

$$30 \leq \text{rad}(bN) \leq |J| < \frac{21}{10} a \ln a,$$

and the radical contains at least three primes coprime to a . This excludes $2 \leq a \leq 7$, since $\ln 7 < 2$. For $a = 8, 9, 10, 12$, the least possible products of three allowable primes are 105, 70, 231, 385; the bound $\ln a < 3$ excludes each. $\square$

Lemma 31 (Excluding constant term eleven). *No factor in a hypothetical factorization has constant term 11.*

Proof. Suppose $a = 11$. The preceding radical bound gives $\text{rad}(bN) < 70$. It is even, is coprime to 11, and has at least three prime divisors, so it is 30 or 42: the next allowable product of three distinct primes is already $2 \cdot 5 \cdot 7 = 70$. If n is even, these supports force b to be a power of two and $3 \mid N$. Lemma 25 gives even d , so $11^N \equiv 1 \pmod{3}$ contradicts odd N .

Thus b is odd. For its least prime divisor q , the order of 11 modulo q (the least positive exponent giving residue one) divides 11: it divides $11b$ by (15), and divides $q - 1$ by Fermat's theorem. Every prime divisor of b is at least q , so $\gcd(b, q - 1) = 1$. Order 11 would give $q \geq 23$, outside the possible supports; order one forces $q = 5$. Since N is not a prime power,

$$b = 5^u, \quad N = 2^v 3^w, \quad u, v, w \geq 1.$$

Reduce $11 \cdot 5^u + 1 = 2^v 3^w$ modulo 3, 8, 5, successively, to obtain u even, $v = 2$, and $w \equiv 2 \pmod{4}$. Indeed, modulo three we get $(-1)^{u+1} + 1 = 0$; then $5^u \equiv 1$ modulo eight gives $11 \cdot 5^u + 1 \equiv 4$, so $v = 2$. Modulo five, $4 \cdot 3^w \equiv 1$ forces $w \equiv 2 \pmod{4}$. Hence

$$(2 \cdot 3^{w/2} - 1)(2 \cdot 3^{w/2} + 1) = 11 \cdot 5^u.$$

The two left factors are coprime odd integers greater than one. They must be 11 and 5^u ; their difference two would force $5^u = 9$ or 13, which is impossible. $\square$

Proposition 32 (Completion of the arithmetic exclusions). *Every hypothetical nontrivial monic integer factorization has $a, b \geq 13$. In particular, no such factorization exists for $n < 182$.*

Proof. Apply Lemmas 30 and 31 to each factor. Then $n = ab \geq 13 \cdot 14 = 182$ by coprimality. $\square$

Proof of Theorem 1. By Gauss's lemma, a reducible monic f_n would admit a nontrivial monic integer factorization. Proposition 32 forces both constant terms to be at least 13. Proposition 24 rules out precisely such a factorization. Thus f_n is irreducible over $\mathbb{Q}$, and monicity gives irreducibility in $\mathbb{Z}[x]$ as well. $\square$